

The dual simplex method

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We consider a linear program in standard form:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \\ & x \geq 0_n, \end{aligned} \quad (\text{P})$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$. We assume $\text{rank}(A) = m$. The set $N := \{1, \dots, n\}$ is used to index the variables. The j^{th} column of A is denoted A_j . The dual linear program (D) of (P) is given by:

$$\begin{aligned} \max \quad & b^T y \\ \text{s.t.} \quad & A^T y \leq c, \end{aligned} \quad (\text{D})$$

and $F_D := \{y \in \mathbb{R}^m : A^T y \leq c\}$ denotes the polyhedron of feasible solutions to (D).

1 Bases and basis matrices

A *basis* of A is an m -subset $B \subseteq N$ such that the columns $\{A_i : i \in B\}$ are linearly independent. Two bases of A are *adjacent* if they only differ by one element.

Given an ordering of the basic columns $\{A_i : i \in B\}$ one has a *basis matrix*. Specifically, given a fixed ordering $(i_1, \dots, i_m)$ of the indices in $B = \{i_1, \dots, i_m\}$, the associated basis matrix is $A_B = [A_{i_1}, \dots, A_{i_m}]$. A basis matrix A_B is non-singular since the columns $\{A_i : i \in B\}$ are linearly independent.

The matrix $\bar{A} := A_B^{-1} A$ is called the *updated A matrix*. The j^{th} column of $\bar{A}$ is denoted $\bar{A}_j$. The non-zero entries of $\bar{A}$ characterize the adjacent bases of B :

Observation 1 (*Characterization of adjacent bases of A*)

Let $B' := (B \cup \{j\}) \setminus \{i_k\}$ be the set obtained by replacing the basic column $i_k \in B$ with the non-basic column $j \in N \setminus B$. The set B' is a basis of $A \iff \bar{a}_{k,j} \neq 0$. ■

Indeed, the vectors $\{A_i : i \in B'\}$ are linearly independent $\iff$ the vectors $\{\bar{A}_i : i \in B'\}$ are linearly independent, and the vectors $\{\bar{A}_i : i \in B\}$ are the unit vectors in $\mathbb{R}^m$. Hence the vectors $\{\bar{A}_i : B \setminus \{i_k\}\}$ all have zero on the k^{th} coordinate. It follows that the set $B' = (B \cup \{j\}) \setminus \{i_k\}$ is a basis of $A \iff \bar{A}_j \notin \text{Span}(\{\bar{A}_i : i \in B \setminus \{i_k\}\})$ holds $\iff \bar{a}_{k,j} \neq 0$.

2 Basic solutions

The *primal basic solution* associated with B is the unique solution $\bar{x} \in \mathbb{R}^n$ to the system:

$$Ax = b \quad \text{and} \quad x_j = 0 \quad \text{for } j \in N \setminus B,$$

and the *dual basic solution* associated with B is the unique solution $\bar{y} \in \mathbb{R}^m$ to:

$$A_i^T \bar{y} = c_i \quad \text{for all } i \in B.$$

The primal basic solution $\bar{x}$ associated with B has $\bar{x}_j = 0$ on non-basic coordinates $j \in N \setminus B$, and the basic coordinates are obtained by solving the remaining system with m variables and m equalities:

$$A_B \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = b \iff \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = A_B^{-1} b$$

A formula can also be given for the dual basic solution associated with B . Let $c^B \in \mathbb{R}^m$ be the vector of cost coefficients associated with basic variables only:

$$c^B := \begin{pmatrix} c_{i_1} \\ \vdots \\ c_{i_m} \end{pmatrix}.$$

The dual basic solution associated with B can be then calculated by solving a system of m equalities and m variables:

$$A_i^T \bar{y} = c_i \text{ for } i \in B \iff \bar{y}^T A_B = (c^B)^T \iff \bar{y}^T = (c^B)^T A_B^{-1}.$$

If the dual basic solution $\bar{y}$ associated with B satisfies $A_j^T \bar{y} \leq c_j$ for all $j \in N$, then $\bar{y}$ is called *basic feasible* for (D), and if the primal basic solution $\bar{x}$ associated with B satisfies $\bar{x}_j \geq 0$ for all $j \in N$, then $\bar{x}$ is called *basic feasible* for (P).

The primal and dual basic solutions $\bar{x}$ and $\bar{y}$ (not necessarily feasible) have the same objective function value in their respective problems:

$$\bar{y}^T b = (c^B)^T A_B^{-1} b = (c^B)^T \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = c^T \bar{x}.$$

From weak duality it then follows that, if $\bar{x}$ is primal feasible, and if $\bar{y}$ is dual feasible, then $\bar{x}$ is optimal for (P), and $\bar{y}$ is optimal for (D).

3 A translated cone associated with every basis

We now consider a polyhedron associated with a basis B :

$$\begin{aligned} F_D(B) &:= \{y \in \mathbb{R}^m : A_i^T y \leq c_i \text{ for } i \in B\} \\ &= \{y \in \mathbb{R}^m : A_B^T y \leq (c^B)\}. \end{aligned}$$

The polyhedron $F_D(B)$ is obtained from F_D by eliminating inequalities corresponding to non-basic columns of A . Hence $F_D(B)$ is a relaxation of F_D , *i.e.*, one has $F_D \subseteq F_D(B)$. The polyhedron $F_D(B)$ gives relevant "local" information" at $\bar{y}$: All inequalities that are not tight at $\bar{y}$ are removed from the description.

The polyhedron $F_D(B)$ has exactly one basic feasible solution, namely the dual basic solution $\bar{y}$ obtained by setting *all* inequalities of $F_D(B)$ to equality. Hence $F_D(B)$ is a translated cone, *i.e.*, one has $F_D(B) = \bar{y} + \text{rec}(F_D(B))$, where $\text{rec}(F_D(B))$ is the recession cone of $F_D(B)$:

$$\begin{aligned} \text{rec}(F_D(B)) &= \{y \in \mathbb{R}^m : A_B^T y \leq 0_m\}. \\ &= \{y \in \mathbb{R}^m : A_B^T y + s = 0_m \text{ for some } s \geq 0_m\}. \\ &= \{y \in \mathbb{R}^m : y = -(A_B^{-1})^T s \text{ for some } s \geq 0_m\}. \\ &= \text{cone}(\{-\bar{b}_1, \dots, -\bar{b}_m\}), \end{aligned}$$

where:

$$\bar{b}_k \sim k^{\text{th}} \text{ row of } A_B^{-1} \quad \text{for } k \in \{1, \dots, m\}.$$

We conclude that one has the following description of $F_D(B)$ in terms of extreme points and extreme directions:

Lemma 1 (*Description of the polyhedron $F_D(B)$*)
The polyhedron $F_D(B)$ is given by:

$$F_D(B) = \bar{y} + \text{cone}(\{-\bar{b}_1, \dots, -\bar{b}_m\}),$$

where $\bar{b}_1, \dots, \bar{b}_m$ are the rows of A_B^{-1} . ■

4 The dual simplex method

The dual simplex method is an iterative method that starts with a dual feasible basis B .

It is first tested whether the information available at B allows one to conclude that (D) is unbounded, or that the associated basic solution $\bar{y}$ is optimal for (D). If this is possible, the dual simplex method terminates giving this information.

If not, the dual simplex method attempts to identify an adjacent dual feasible basis B' whose associated basic solution has better dual objective value than $\bar{y}$. The dual simplex method is then iterated with B' as the new starting basis.

Hence it suffices to describe a single iteration of the dual simplex method. Therefore suppose B is a dual feasible basis, let $\bar{y}$ be the corresponding basic feasible solution for (D), and let $\bar{x}$ be the corresponding basic solution for (P).

The vector $\bar{s} := c - A^T \bar{y}$ denotes the slack of $\bar{y}$ in the inequalities $A^T y \leq c$. One has $\bar{s} \geq 0_n$ since $\bar{y}$ is assumed to be feasible for (D). The dual slack variables are also called the *reduced costs* associated with the basis B .

If $\bar{x}_{i_k} \geq 0$ holds for all $k \in \{1, \dots, m\}$, then $\bar{x}$ is basic feasible for (P). Therefore, since we assumed $\bar{y}$ is basic feasible for (D), and since $\bar{x}$ and $\bar{y}$ have the same objective value in their respective problems, weak duality shows $\bar{x}$ and $\bar{y}$ are both optimal solutions. Hence the dual simplex method terminates if $\bar{x}_{i_k} \geq 0$ holds for all $k \in \{1, \dots, m\}$.

Otherwise a basic variable $i_{k_*} \in B$ satisfying $\bar{x}_{i_{k_*}} < 0$ is chosen, where $k_* \in \{1, \dots, m\}$. The dual simplex method searches for a better dual feasible solution than $\bar{y}$ along the halfline that starts at $\bar{y}$ and moves in the extreme direction $-\bar{b}_{k_*}$ of $\text{rec}(F_D(B))$, i.e., along the halfline:

$$H^* := \{y(\theta) : \theta \geq 0\},$$

where $y(\theta) := \bar{y} + \theta \cdot (-\bar{b}_{k_*})$.

All points on the halfline H^* (except $\bar{y}$) have better objective value than $\bar{y}$. Indeed, for any point $y(\theta)$ on H^* with $\theta > 0$ one has:

$$b^T y(\theta) = b^T (\bar{y} - \theta \cdot \bar{b}_{k_*}) = b^T \bar{y} - \theta \cdot \bar{x}_{i_{k_*}} > b^T \bar{y}.$$

where $\bar{x}_{i_{k_*}} = (\bar{b}_{k_*})^T b$ holds since $\bar{x}_{i_{k_*}}$ is the k_* th coordinate of $A_B^{-1} b$.

The dual simplex method tries to improve the dual objective value by moving as far as possible along the halfline H^* while preserving dual feasibility, i.e., the dual simplex method computes the value:

$$\theta^* = \sup\{\theta : A_j^T y(\theta) \leq c_j \text{ for all } j \in N\}.$$

The inequalities $A_i^T y(\theta) \leq c_i$ are satisfied for all $i \in B$ and $\theta \geq 0$. Indeed, one has $A_B^T \bar{b}_{k_*} = e_{k_*}$ since $\bar{b}_{k_*}$ is the k_* th row of A_B^{-1} , and $A_B^T \bar{y} = c^B$ holds since $\bar{y}$ is the dual basic solution associated with B , and therefore:

$$A_B^T y(\theta) = A_B^T (\bar{y} - \theta \cdot \bar{b}_{k_*}) = c^B - \theta \cdot e_{k_*} \leq c^B \quad \text{for all } \theta \geq 0. \quad (1)$$

Hence only the inequalities $A_j^T y(\theta) \leq c_j$ for $j \in N \setminus B$ limit the size of θ . One can now give a formula for θ^* . Observe that $\bar{b}_{k_*}^T A_j = \bar{a}_{k_*,j}$ holds for all $j \in N \setminus B$. Hence :

$$\begin{aligned} \theta^* &= \sup\{\theta : A_j^T y(\theta) \leq c_j \text{ for all } j \in N \setminus B\} \\ &= \sup\{\theta : \bar{s}_j + \theta \cdot \bar{a}_{k_*,j} \geq 0 \text{ for all } j \in N \setminus B\} \\ &= \inf\left\{-\frac{\bar{s}_j}{\bar{a}_{k_*,j}} : \text{for all } j \in N \setminus B \text{ satisfying } \bar{a}_{k_*,j} < 0\right\}. \end{aligned} \quad (2)$$

If $\theta^* = +\infty$ (which is the case when $\bar{a}_{k^*,j} \geq 0$ holds for all $j \in N \setminus B$), then $y(\theta)$ is feasible for (D) for all $\theta \geq 0$, and therefore (D) is unbounded. In this case the dual simplex method terminates.

If $\theta^* < +\infty$, then $\theta^* = -\frac{\bar{s}_{j^*}}{\bar{a}_{k^*,j^*}}$ holds for some non-basic $j^* \in N \setminus B$ that satisfies $\bar{a}_{k^*,j^*} < 0$. Since $\bar{a}_{k^*,j^*} \neq 0$, Observation 1 shows that the set:

$$B' := (B \cup \{j^*\}) \setminus \{i_{k^*}\}$$

is an adjacent basis to B .

Also, the vector $y(\theta^*)$ is the dual basic feasible solution associated with B' . Indeed, from (1) one has $A_B^T y(\theta^*) = c^B - \theta^* \cdot e_{k^*}$, and therefore $A_i^T y(\theta^*) = c_i$ holds for all $i \in B \setminus \{i_{k^*}\}$. Finally the formula for θ^* gives:

$$\begin{aligned} A_{j^*}^T y(\theta^*) &= A_{j^*}^T (\bar{y} - \theta^* \cdot \bar{b}_{k^*}) && \text{(from the definition of } y(\theta^*)) \\ &= A_{j^*}^T \bar{y} - \left(-\frac{\bar{s}_{j^*}}{\bar{a}_{k^*,j^*}}\right) \cdot A_{j^*}^T \bar{b}_{k^*} && \text{(inserting the formula for } \theta^*) \\ &= A_{j^*}^T \bar{y} + \bar{s}_{j^*} = c_{j^*}. && \text{(since } \bar{a}_{k^*,j^*} = A_{j^*}^T \bar{b}_{k^*} \text{ and } \bar{s}_{j^*} = c_{j^*} - A_{j^*}^T \bar{y}) \end{aligned}$$

Hence $A_i^T y(\theta^*) = c_i$ holds for all $i \in B'$.

We conclude that B' is a dual feasible basis with $y(\theta^*)$ as the associated basic solution, and that $y(\theta^*)$ has better dual objective value than $\bar{y}$ when $\theta^* > 0$:

Observation 2 (*An adjacent basis*)

The basis B' is dual feasible and adjacent to B , and $y(\theta^)$ is the corresponding basic solution. Also, if $\theta^* > 0$, then $y(\theta^*)$ has better dual objective value than $\bar{y}$. ■*

If it has not been discovered that (D) is unbounded, or that $\bar{y}$ is optimal for (D), the dual simplex method performs another iteration with B' as the new starting basis. The procedure is continued until either an optimal solution has been found, or it has been discovered that (D) is unbounded.

There is no guarantee of improvement in every iteration, *i.e.*, one may have $\theta^* = 0$. However, this turns out not to be a big problem in practice.

5 Example

We now consider an example with $n = 5$ variables, so $N = \{1, \dots, 5\}$. There are 2 equality constraints. The problems (P) and (D) are given by:

$$\begin{aligned} \min \quad & c^T x & \max \quad & b^T y \\ \text{s.t.} \quad & & \text{s.t.} \quad & \\ & Ax = b & & A^T y \leq c & (P) & (D) \\ & x \geq 0_n & & & \end{aligned}$$

where:

$$A = \begin{pmatrix} -2 & 4 & 1 & 1 & 0 \\ 4 & -2 & -3 & 0 & 1 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 2 \\ -1 \end{pmatrix},$$

and:

$$c^T = (2 \quad 6 \quad 10 \quad 0 \quad 0).$$

We consider the basis $B = \{4, 5\}$ with ordering $(i_1, i_2) = (4, 5)$. Hence $A_B = [A_4, A_5] = I_2 = A_B^{-1}$ and $c^B = 0_2$.

The primal basic solution $\bar{x} \in \mathbb{R}^5$ solves $\bar{x}_j = 0$ for the non-basic variables $j \in N \setminus B = \{1, 2, 3\}$, and $\begin{pmatrix} \bar{x}_{i_1} \\ \bar{x}_{i_2} \end{pmatrix} = \begin{pmatrix} \bar{x}_4 \\ \bar{x}_5 \end{pmatrix} = A_B^{-1}b = b = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Hence:

$$\bar{x}^T = (0 \quad 0 \quad 0 \quad 2 \quad -1),$$

which is *not* feasible for (P) since $\bar{x}_5 < 0$.

The dual basic solution associated with B is given by $\bar{y}^T = (c^B)^T A_B^{-1} = (0, 0)$, and the slacks $\bar{s}$ in the dual constraints are given by $\bar{s} = c - A^T \bar{y} = c$. Hence:

$$\bar{y}^T = (0 \quad 0) \quad \text{and} \quad \bar{s}^T = (2 \quad 6 \quad 10 \quad 0 \quad 0).$$

which shows B is a dual feasible basis.

We have $A_B^{-1} = I_2$, so the two rows of A_B^{-1} are given by:

$$\bar{b}_1^T = (1 \quad 0) \quad \text{and} \quad \bar{b}_2^T = (0 \quad 1),$$

and also $\bar{A} = A_B^{-1}A = A$.

We first choose a variable to exit the basis, *i.e.*, a basic variable with negative value in the basic solution. We have $\bar{x}_{i_1} = \bar{x}_4 = 2$ and $\bar{x}_{i_2} = \bar{x}_5 = -1$. Hence the dual simplex method chooses x_5 to leave the basis. We have that x_5 is the second basic variable, *i.e.*, one has $i_2 = 5$ and $k_* = 2$.

The entering non-basic variable is determined by computing the value θ^* :

$$\begin{aligned} \theta^* &= \inf \left\{ -\frac{\bar{s}_j}{\bar{a}_{k_*,j}} : \text{for all } j \in N \setminus B \text{ satisfying } \bar{a}_{k_*,j} < 0 \right\}. \\ &= \inf \left\{ -\frac{\bar{s}_2}{\bar{a}_{2,2}}, -\frac{\bar{s}_3}{\bar{a}_{2,3}} \right\} = \inf \left\{ -\frac{6}{(-2)}, -\frac{10}{(-3)} \right\} = 3. \end{aligned}$$

The minimum was attained for $j_* = 2$. Hence x_2 enters the basis and x_5 leaves the basis, which gives the basis B' :

$$B' := (B \setminus \{5\}) \cup \{2\} = \{2, 4\}$$

with new ordering $(i'_1, i'_2) = (4, 2)$ (since $i_2 = 5$ exits the basis).

The new dual basic feasible solution $y(\theta^*)$ is given by:

$$y(\theta^*) = \bar{y} + 3(-\bar{b}_{k_*}) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$$

which has a dual objective $b^T \bar{y}' = 3$. This is better than $\bar{y}$, since $\theta^* = 3 > 0$. The new primal basic solution from $B' = \{2, 4\}$ is given by:

$$(x(\theta^*))^T = \left(0 \quad \frac{1}{2} \quad 0 \quad 0 \quad 0\right).$$

Observe that, since $x(\theta^*)$ is feasible for (P) *and* $y(\theta^*)$ is feasible for (D), then $\bar{x}'$ is optimal for (P), and $y(\theta^*)$ is optimal for (D).